Periodic and Oscillatory Motion

Class II · x measured from EQUILIBRIUM · ideal MASSLESS springs, FRICTIONLESS surfaces, uniform g unless stated · ω = angular frequency of the undamped SHM, ω_d = driving

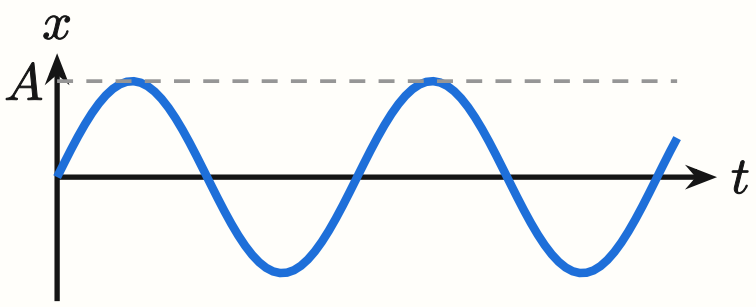
1 Periodic motion

$$T = \frac{1}{f} \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

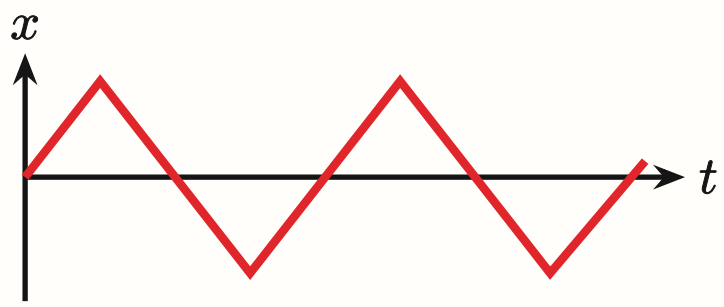
REPEATS after a fixed time T (s) · f in Hz · ω in $\mathrm{rad\,s^{-1}}$

- *Oscillatory: to and fro about an EQUILIBRIUM position. Every UNDAMPED oscillation is periodic; uniform circular motion is periodic, not oscillatory.*

2 Oscillatory, but is it SHM?



SHM: $x = A\sin\omega t$



periodic, oscillatory, NOT SHM

3 Which functions?

- *$\sin\omega t + \cos\omega t = \sqrt{2}\sin(\omega t + \frac{\pi}{4})$: SHM, amplitude $\sqrt{2}$, period $\frac{2\pi}{\omega}$.*
- *$\sin^2\omega t = \frac{1}{2} - \frac{1}{2}\cos 2\omega t$: SHM about $x = \frac{1}{2}$, period $\frac{\pi}{\omega}$.*
- *$\sin\omega t + \sin 2\omega t$: periodic (period $\frac{2\pi}{\omega}$) but NOT SHM. $e^{-\omega t}$: not periodic.*

Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

1 JEE Main + JEE Advanced + NEET syllabus

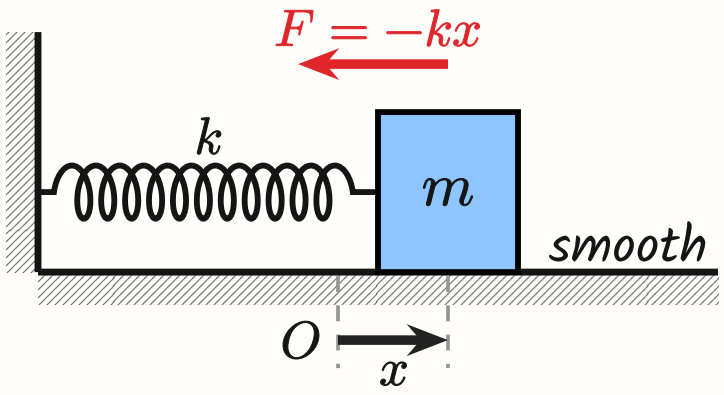
- ✓ Periodic motion: period, frequency — p1
- ✓ SHM: $F = -kx$, $a = -\omega^2 x$; equations, phase — p3, p4
- ✓ Phase relations, x, v, a graphs; v–x, a–x — p5–p7
- ✓ Energy in SHM: KE, PE, total; graphs; averages — p8, p9
- ✓ Spring–mass: horizontal, vertical, series, parallel — p10, p11
- ✓ Spring cut into parts; massive spring correction — p12
- ✓ Simple pendulum (small angle); g_{eff} in a lift, cart, fluid — p13, p14, p16
- ✓ Physical (compound) and torsional pendulum — p17, p18
- ✓ Damped oscillations; forced oscillations, resonance — p25–p27
- ✓ Pendulum tension; seconds pendulum; large amplitude — p13

2 Also covered: systems + standard results

- ✓ Reference circle; time between two positions — p6
- ✓ Seconds pendulum; very long pendulum ($l \sim R$) — p13, p15
- ✓ U-tube liquid; floating body; gas column, piston — p19, p20
- ✓ Tunnel through the earth — p21
- ✓ SHM from $U(x)$: $\omega^2 = U''(x_0)/m$ — p22
- ✓ Superposition: same direction, perpendicular — p23, p24
- ✓ Two bodies on a spring (reduced mass) — p12
- ✓ Angular SHM: restoring torque $\tau = -\kappa\theta$ — p3, p17, p18
- ✓ Special cases, results, traps, solved questions — p28–p37

SHM: Definition

1 Restoring force



$$F = -kx \qquad a = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

x from EQUILIBRIUM O · k = force constant ($\mathrm{N\,m^{-1}}$) · ideal massless spring, frictionless surface

2 Definition

- *SHM: acceleration (or restoring force) is proportional to the displacement from equilibrium and always directed TOWARDS it.*
- *Equation of motion:* $\dfrac{d^2x}{dt^2} + \omega^2 x = 0$. *Only a LINEAR restoring force gives SHM;* $F = -kx^3$ *oscillates, but it is not SHM.*
- *Angular SHM (rigid body about a fixed axis):* $\tau = -\kappa\theta$ *gives* $\alpha = -\omega^2\theta$ *with* $\omega^2 = \kappa/I$.

3 Reading ω off an equation

Example

$a = -16x$ *(SI units):* $\omega^2 = 16$, $\omega = 4\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{4} = 1.57\,\mathrm{s}$.

$a = -16x + 8$: *SHM about* $x_0 = 0.5\,\mathrm{m}$ *(where* $a = 0$*), same* $\omega = 4\,\mathrm{rad\,s^{-1}}$.

Equations of SHM

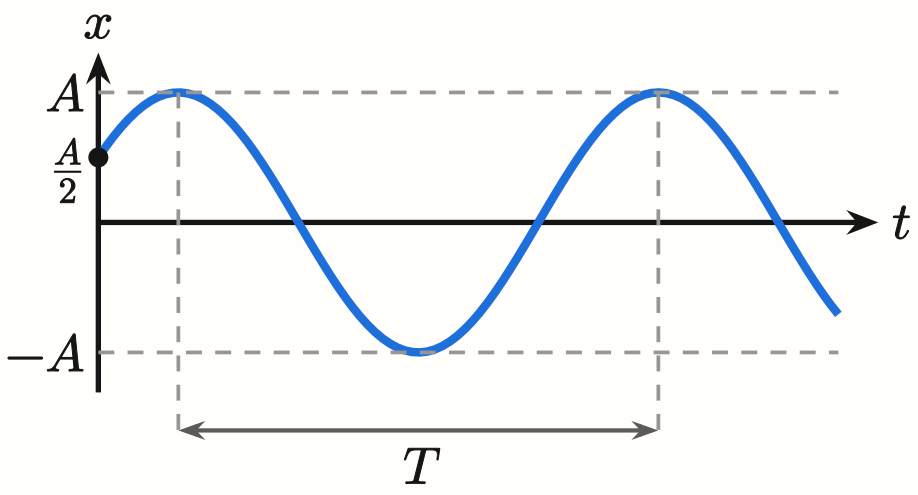
1 Displacement, velocity, acceleration

$$x = A\sin(\omega t + \phi)$$

$$v = A\omega\cos(\omega t + \phi) \qquad a = -A\omega^2\sin(\omega t + \phi)$$

$a = -\omega^2 x$ · x from equilibrium · A = amplitude (maximum $|x|$) · $(\omega t + \phi)$ = PHASE (rad), fixes position AND direction · ϕ = initial phase

2 The graph



- Drawn: $\phi = \frac{\pi}{6}$, so $x(0) = \frac{A}{2}$, moving in $+x$.
- $\phi = 0$: at O, moving $+x$. $\phi = \frac{\pi}{2}$: at $+A$. $\phi = \pi$: at O, moving $-x$.

3 Amplitude and ϕ from the start

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \qquad \tan\phi = \frac{\omega x_0}{v_0}$$

x_0, v_0 at $t = 0$ · for $x = A\sin(\omega t + \phi)$ · choose the quadrant from the SIGNS: $\sin\phi = \frac{x_0}{A}$, $\cos\phi = \frac{v_0}{A\omega}$

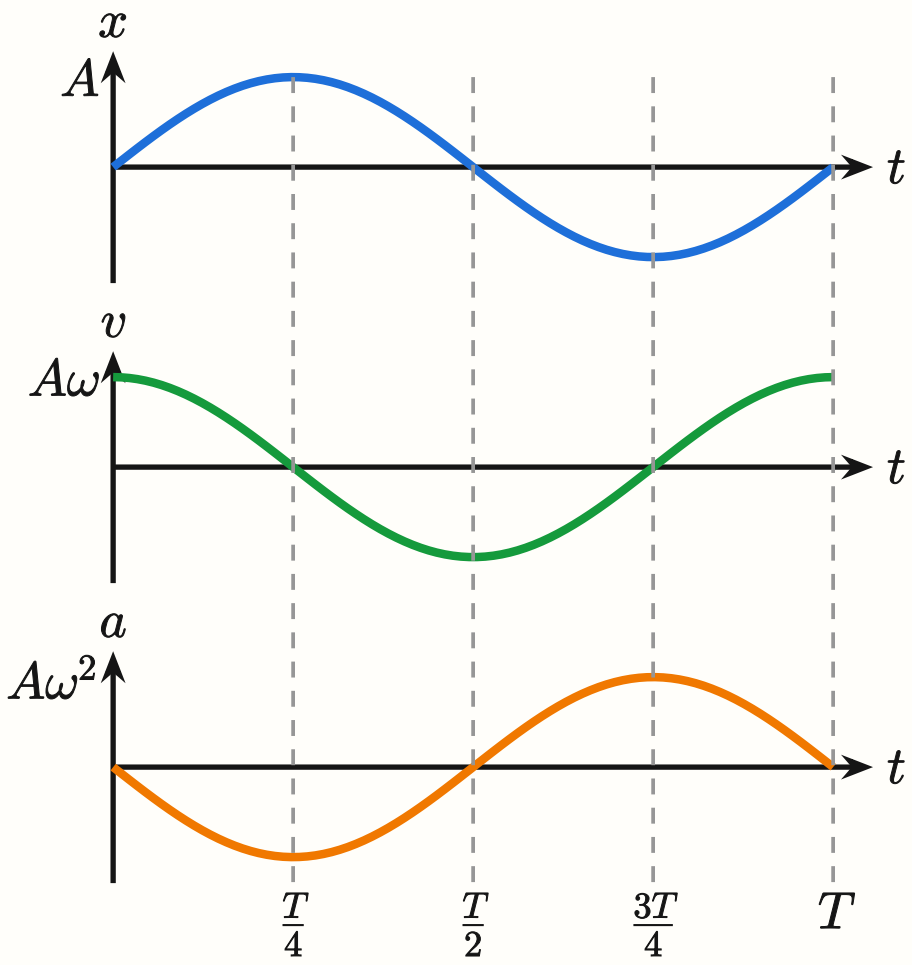
Example

$\omega = 2\,\text{rad}\,\text{s}^{-1}$, $x_0 = 3\,\text{cm}$, $v_0 = 8\,\text{cm}\,\text{s}^{-1}$: $A = \sqrt{9 + 16} = 5\,\text{cm}$; $\sin\phi = 0.6$, $\cos\phi = 0.8$, $\phi = 37°$.

Phase Relations and Graphs

$x = A\sin\omega t$ (initial phase zero) for the graphs

1 x, v, a against t



v leads x by $\frac{\pi}{2}$

a leads v by $\frac{\pi}{2}$

a and x : phase π

same ω for all three · $a = -\omega^2 x$ always

2 Reading the graphs

- At the MEAN position: $|v| = A\omega$ (max), $a = 0$. At the EXTREMES: $v = 0$, $|a| = A\omega^2$ (max).
- Slope of the x–t graph $= v$; slope of the v–t graph $= a$.

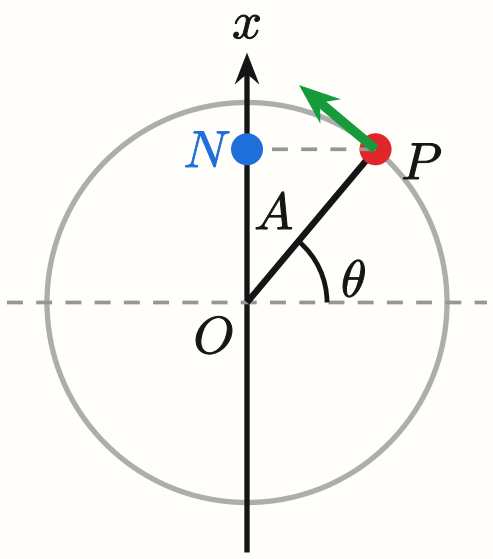
3 Phase difference of two particles

Example

Two particles, same A and ω, cross at $x = \frac{A}{2}$ moving in OPPOSITE directions: phases $\frac{\pi}{6}$ and $\frac{5\pi}{6}$, phase difference $\frac{2\pi}{3}$.

Reference Circle and Time

1 SHM = projection of uniform circular motion



$$ON = A\sin\theta \qquad \theta = \omega t + \phi$$

P: circle of radius A, CONSTANT ω · its projection N on a diameter performs SHM

2 Time between positions

$$0 \to \frac{A}{2} : \frac{T}{12} \qquad \frac{A}{2} \to A : \frac{T}{6}$$

$$0 \to \frac{A}{\sqrt{2}} : \frac{T}{8} \qquad 0 \to \frac{\sqrt{3}A}{2} : \frac{T}{6}$$

along the direct path, no turning back · $\Delta t = \Delta(\text{phase})/\omega$

Example

$T = 2.4\,\text{s}$, $x = \frac{A}{2}$ *through* O *to* $-\frac{A}{2}$: $2 \times \frac{T}{12} = 0.4\,\text{s}$.

3 Distance

- *Distance:* $4A$ *per period,* $2A$ *per half period from ANY start;* A *in* $\frac{T}{4}$ *only from* O *or an extreme.*
- *Average speed over a period* $= \frac{4A}{T} = \frac{2A\omega}{\pi}$; *average velocity over a period* $= 0$.

Velocity and Acceleration vs x

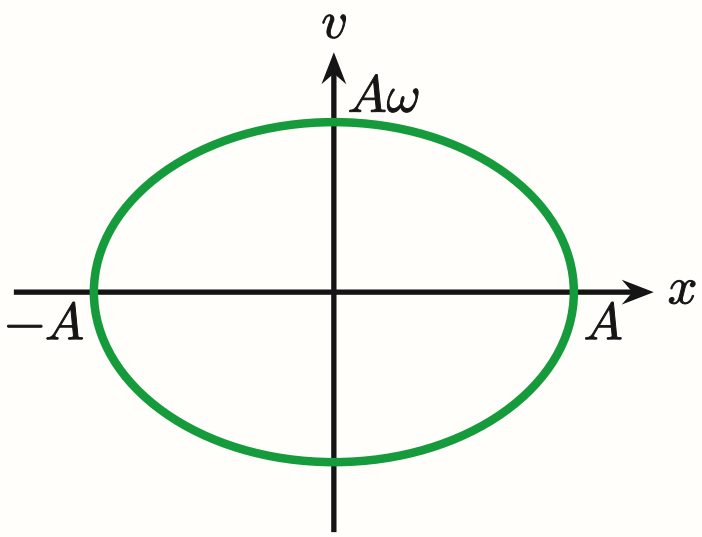
1 v and a in terms of x

$$v = \pm\omega\sqrt{A^2 - x^2} \qquad a = -\omega^2 x$$

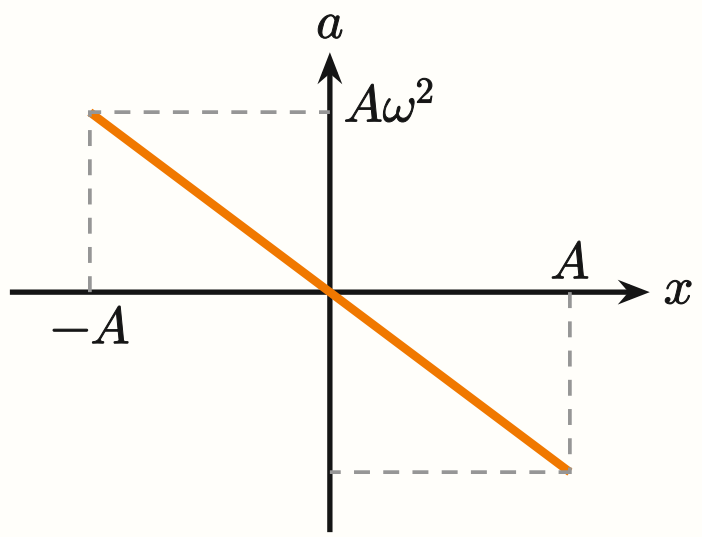
$$v_{max} = A\omega \quad (x = 0) \qquad a_{max} = A\omega^2 \quad (x = \pm A)$$

x from equilibrium · sign of v = direction of motion · from $v^2 = \omega^2(A^2 - x^2)$

2 Graphs



ellipse $\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1$



straight line, slope $-\omega^2$

3 Speeds at two positions

$$\omega^2 = \frac{v_1^2 - v_2^2}{x_2^2 - x_1^2} \qquad A^2 = \frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}$$

speed v_1 at x_1, speed v_2 at x_2, same SHM

Example

$4\,\mathrm{cm\,s^{-1}}$ *at* $3\,\mathrm{cm}$, $3\,\mathrm{cm\,s^{-1}}$ *at* $4\,\mathrm{cm}$: $\omega^2 = \frac{16-9}{16-9} = 1$, $\omega = 1\,\mathrm{rad\,s^{-1}}$;
$A^2 = \frac{256-81}{7} = 25$, $A = 5\,\mathrm{cm}$.

Energy in SHM

PE taken ZERO at the equilibrium position · no friction, no damping

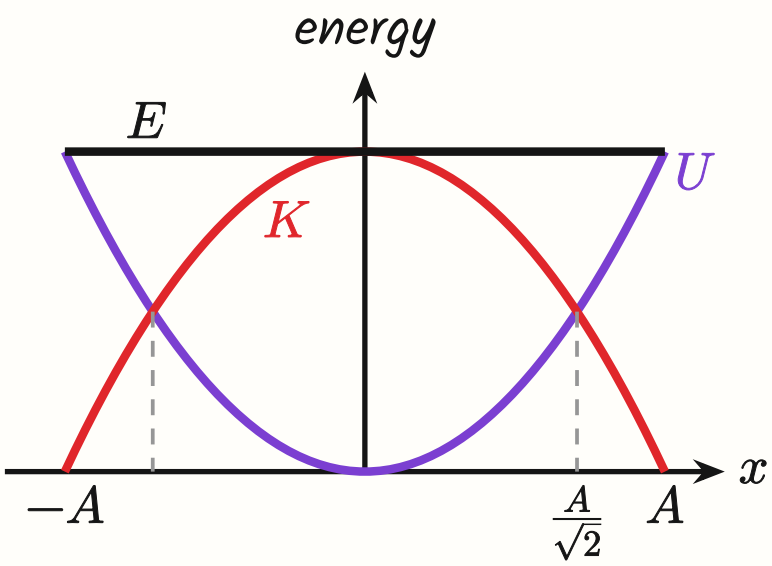
1 Kinetic, potential, total

$$K = \tfrac{1}{2}m\omega^2(A^2 - x^2) \qquad U = \tfrac{1}{2}m\omega^2 x^2$$

$$E = K + U = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$k = m\omega^2$ · E CONSTANT (no friction, no damping) · $E \propto A^2$ and $\propto \omega^2$

2 Energy against x



- Parabolas: U max at the extremes, K max at O.
- $K = U$ at $x = \pm\frac{A}{\sqrt{2}}$.
- At $x = \frac{A}{2}$: $K = \frac{3}{4}E$, $U = \frac{1}{4}E$.

3 Numbers; vertical spring

Example

$m = 0.2\,\text{kg}$, $k = 20\,\text{N m}^{-1}$, $A = 0.1\,\text{m}$: $E = \frac{1}{2} \times 20 \times 0.01 = 0.1\,\text{J}$.
At $x = 0.05\,\text{m}$: $U = 0.025\,\text{J}$, $K = 0.075\,\text{J}$;
$v_{max} = \sqrt{2E/m} = 1\,\text{m s}^{-1}$.

- Vertical spring: measure x from the NEW equilibrium; spring PE + gravity PE together then equal $\frac{1}{2}kx^2 +$ constant, so the same formulas hold.

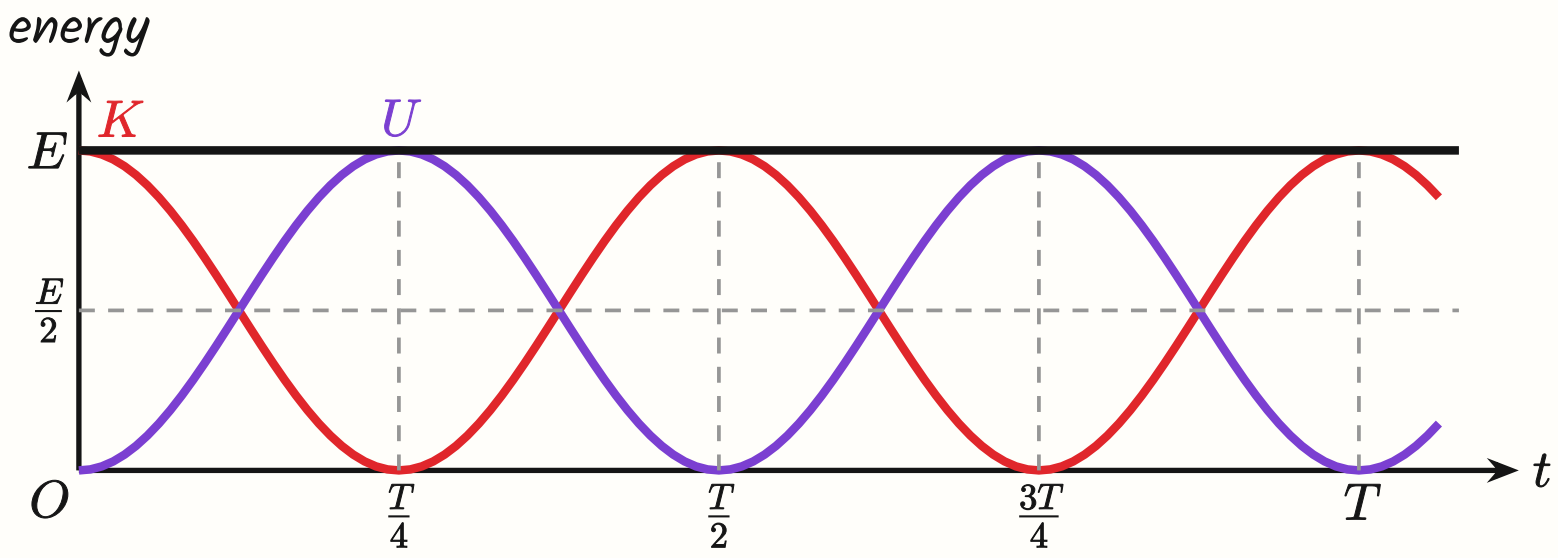
Energy vs Time; Averages

$x = A\sin\omega t$ · **PE zero at equilibrium · no damping**

1 Energy against t

$$K = E\cos^2\omega t \qquad U = E\sin^2\omega t$$

$E = \frac{1}{2}m\omega^2A^2$ · K and U oscillate with period $\frac{T}{2}$ (frequency $2f$) between 0 and E



2 Average values

$$\langle K\rangle_t = \langle U\rangle_t = \frac{E}{2}$$

$$\langle K\rangle_x = \frac{2}{3}E \qquad \langle U\rangle_x = \frac{1}{3}E$$

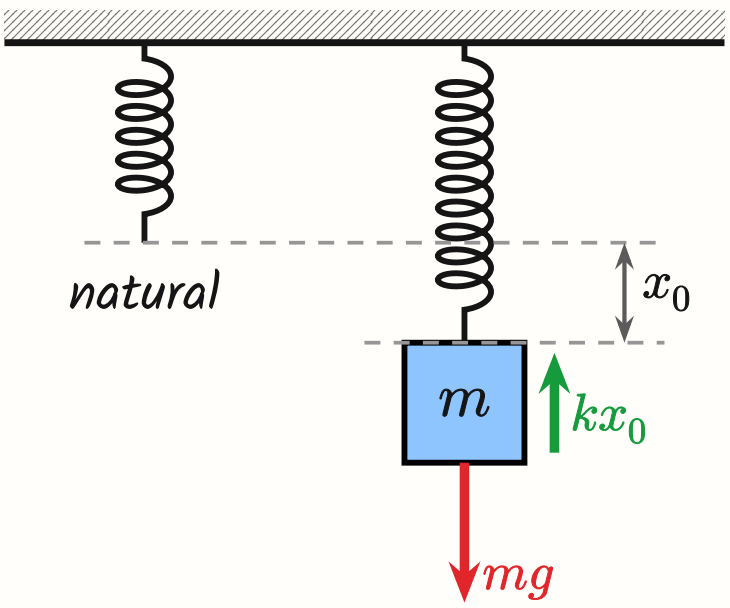
$\langle\ \rangle_t$: averaged over one period in TIME · $\langle\ \rangle_x$: averaged over POSITION, x from $-A$ to A

- $K = U$ first at $t = \frac{T}{8}$ (phase $\frac{\pi}{4}$, $x = \frac{A}{\sqrt{2}}$), then every $\frac{T}{4}$.
- The particle spends MORE time near the extremes (it is slow there), so the time average of U exceeds the position average of U.

Spring-Block Systems

ideal massless spring · rigid support · x from the equilibrium position

1 Horizontal and vertical



$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$x_0 = \frac{mg}{k} \qquad T = 2\pi\sqrt{\frac{x_0}{g}}$$

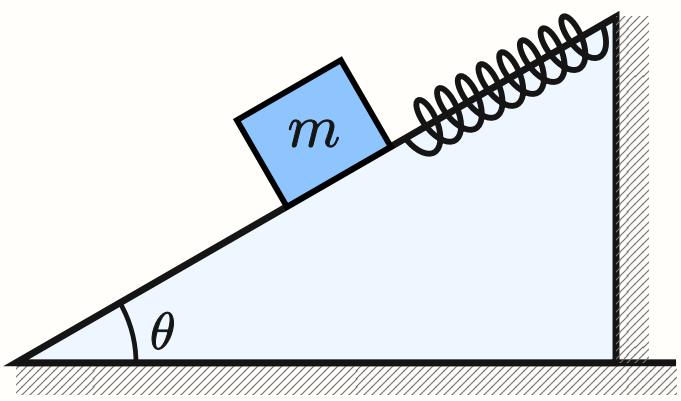
horizontal (smooth floor) OR vertical: SAME T · vertical: SHM about the stretched equilibrium, x_0 = static extension · T independent of A

Example

Static extension $x_0 = 9.8\,\text{cm}$, $g = 9.8\,\text{m s}^{-2}$:

$T = 2\pi\sqrt{0.098/9.8} = 2\pi \times 0.1 = 0.628\,\text{s}.$

2 On a smooth incline



$$T = 2\pi\sqrt{\frac{m}{k}}$$

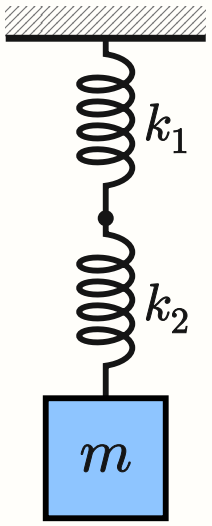
$$x_0 = \frac{mg\sin\theta}{k}$$

smooth incline, spring along the slope, fixed at the top · gravity only SHIFTS the equilibrium, never changes T

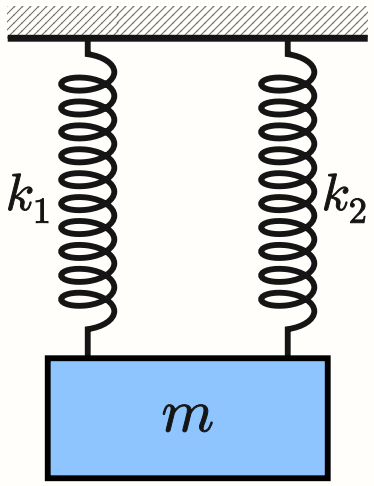
Springs in Series and Parallel

ideal massless springs · smooth floor · rigid supports

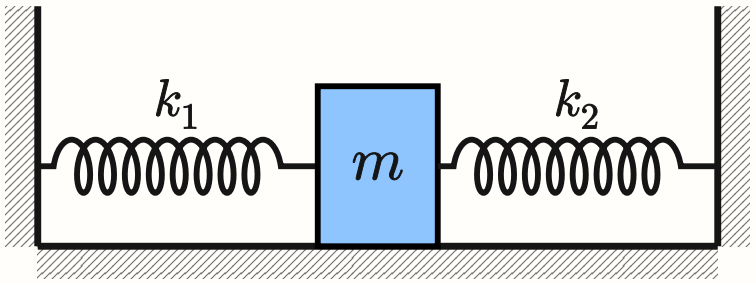
1 The three arrangements



series



parallel



between two walls

2 Effective spring constant

series: $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$ parallel: $k = k_1 + k_2$

block between two walls: $k = k_1 + k_2$

series: SAME force, extensions add · parallel: SAME extension, forces add · between walls: both springs pull the block back the same way: PARALLEL

- Series of two equal springs: $\frac{k}{2}$, T grows $\sqrt{2}$ times. Parallel: $2k$, T falls $\sqrt{2}$ times.

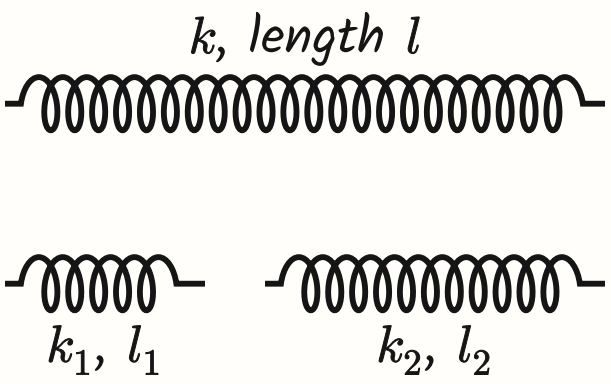
Example

$k_1 = 200\,\mathrm{N\,m^{-1}}$, $k_2 = 300\,\mathrm{N\,m^{-1}}$, $m = 3\,\mathrm{kg}$. Series: $k = 120\,\mathrm{N\,m^{-1}}$, $T = 2\pi\sqrt{3/120} = 0.993\,\mathrm{s}$. Parallel: $k = 500\,\mathrm{N\,m^{-1}}$, $T = 2\pi\sqrt{3/500} = 0.487\,\mathrm{s}$.

Cut Springs, Two Bodies, Massive Spring

uniform spring · smooth floor · small oscillations

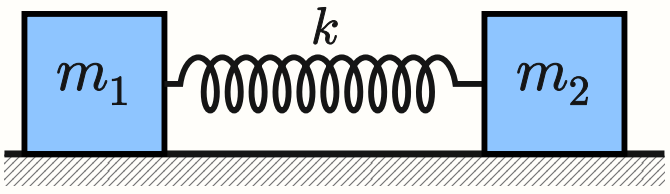
1 Spring cut into parts



$$k\,l = k_1 l_1 = k_2 l_2$$

UNIFORM spring: $k \propto \frac{1}{l}$ · *n equal parts: each* nk; *all* n *in parallel:* $n^2 k$ · $k = 100\,\mathrm{N\,m^{-1}}$ *cut* $1:2$: 300 *and* $150\,\mathrm{N\,m^{-1}}$

2 Two blocks joined by a spring



$$T = 2\pi\sqrt{\frac{\mu}{k}} \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

no external horizontal force (centre of mass at rest) · μ = *reduced mass* · $m_1 = m_2 = 2\,\mathrm{kg}$, $k = 100\,\mathrm{N\,m^{-1}}$: $\mu = 1\,\mathrm{kg}$, $T = 0.628\,\mathrm{s}$

3 Spring with mass

$$T = 2\pi\sqrt{\frac{m + m_s/3}{k}}$$

UNIFORM spring of mass m_s, *one end FIXED, block* m *at the other* · *speed of the coils grows linearly along the spring*

Example

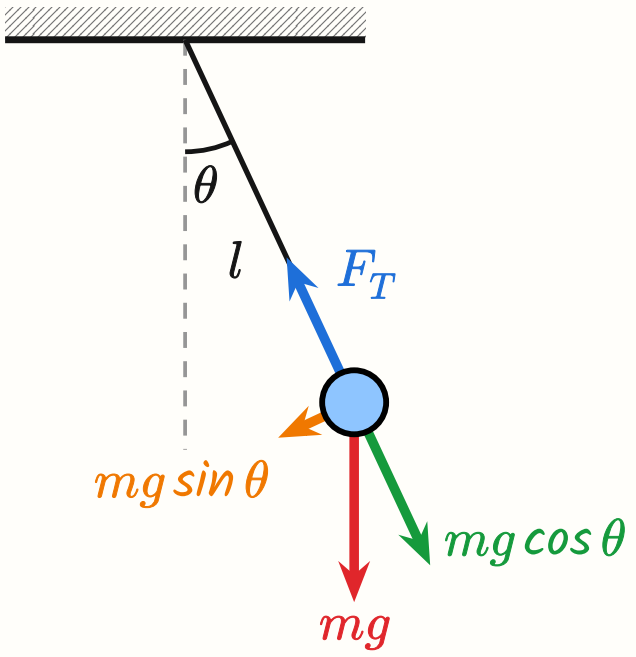
$m = 1\,\mathrm{kg}$, $m_s = 0.3\,\mathrm{kg}$, $k = 110\,\mathrm{N\,m^{-1}}$:

$T = 2\pi\sqrt{1.1/110} = 0.628\,\mathrm{s}.$

Simple Pendulum

point bob, light inextensible string, rigid support, uniform g, no air drag

1 Forces on the bob



$$T = 2\pi\sqrt{\frac{l}{g}}$$

SMALL angle ($\theta_0 \lesssim 10°$, $\sin\theta \approx \theta$) · l = support to the bob's CENTRE · independent of m and of a small amplitude

- Restoring force $mg\sin\theta \approx \frac{mg}{l}s$ (s = arc): linear only for small θ.
- Tension: $F_T = mg\cos\theta + \frac{mv^2}{l}$; maximum at the mean position, $mg(3 - 2\cos\theta_0)$.

Example

$l = 1\,\text{m}$, $g = 9.8\,\text{m}\,\text{s}^{-2}$, small angle: $T = 2\pi\sqrt{1/9.8} = 2.007\,\text{s}$.

2 Seconds pendulum and large amplitude

$$T = 2\,\text{s} \;\Rightarrow\; l = \frac{g}{\pi^2} = 0.993\,\text{m}$$

$$T \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\theta_0^2}{16}\right)$$

$g = 9.8\,\text{m}\,\text{s}^{-2}$ · second line: moderate amplitude θ_0 (rad), first correction only

Pendulum in Accelerating Frames

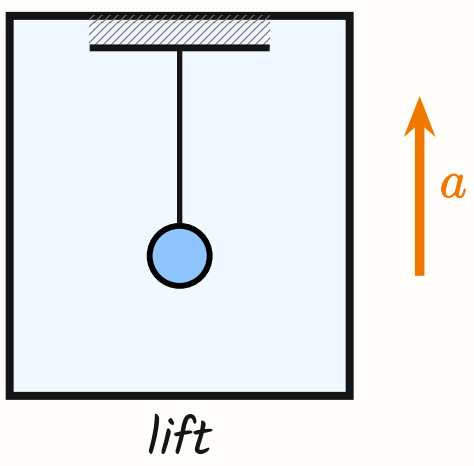
small angle · $\vec{a}$ = acceleration of the support (lift / cart) · uniform g

1 Effective g

$$T = 2\pi\sqrt{\frac{l}{g_{eff}}} \qquad \vec{g}_{eff} = \vec{g} - \vec{a}$$

pseudo force $-m\vec{a}$ in the support's frame · the mean position lies along $\vec{g}_{eff}$

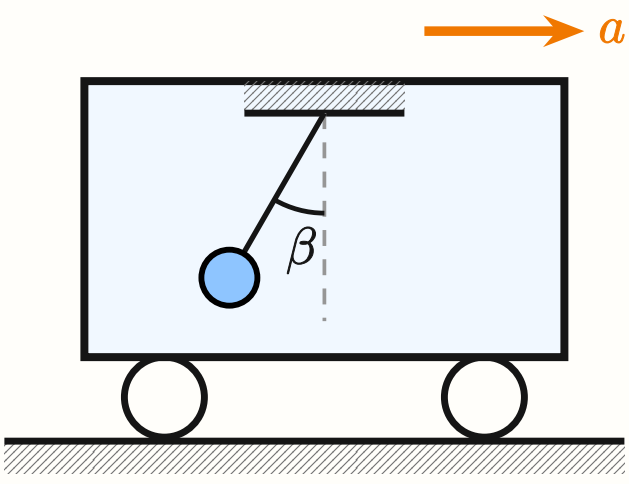
2 Lift



lift

- *Accelerating UP:* $g_{eff} = g + a$, T *smaller.*
- *Accelerating DOWN* ($a < g$): $g_{eff} = g - a$, T *larger. Free fall: no oscillation. Constant velocity:* T *unchanged.*
- $T_0 = 2\,\mathrm{s}$, $a = \frac{g}{4}$ *up:* $T = \frac{2}{\sqrt{1.25}} = 1.789\,\mathrm{s}$.

3 Cart with horizontal acceleration



$$g_{eff} = \sqrt{g^2 + a^2}$$

$$\tan\beta = \frac{a}{g}$$

the string hangs tilted BACK (opposite to $\vec{a}$) by β and oscillates about that line

Very Long Pendulum

small angle · light string, rigid support · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ at the surface

1 Length comparable to the earth's radius

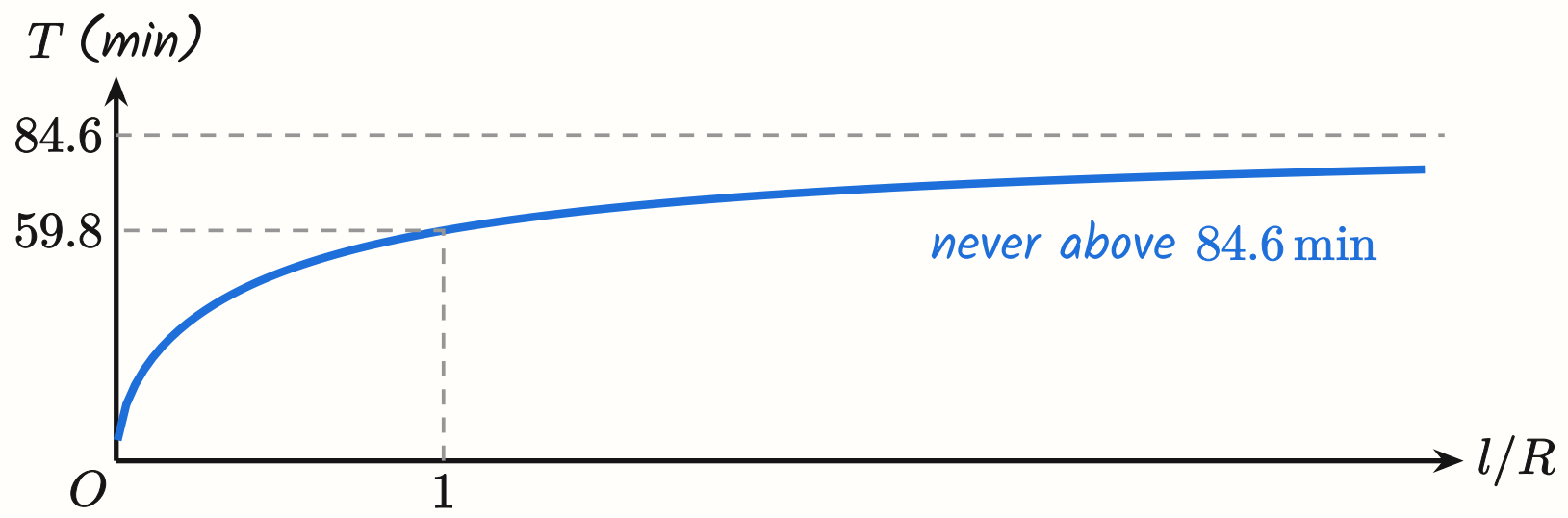
$$T = 2\pi\sqrt{\frac{1}{g\left(\frac{1}{l}+\frac{1}{R}\right)}}$$

$$l \to \infty:\quad T = 2\pi\sqrt{\frac{R}{g}} = 84.6\,\mathrm{min}$$

bob near the surface, g directed to the earth's CENTRE (not parallel) · $l \ll R$ gives back $2\pi\sqrt{l/g}$

- $l = R$: $T = 2\pi\sqrt{\frac{R}{2g}} = 59.8\,\mathrm{min}$.
- *Why: for a long string the restoring force has TWO parts, from the string's tilt ($\propto \frac{s}{l}$) and from $\vec{g}$ turning towards the centre ($\propto \frac{s}{R}$), s = arc moved by the bob.*

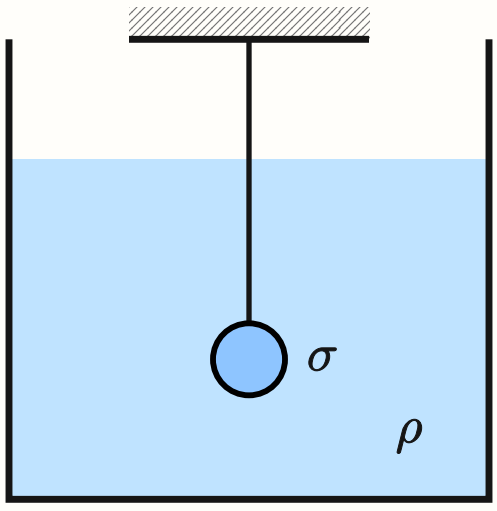
2 Period against length



Pendulum in a Fluid or a Field

small angle · rigid support · uniform g · $T_0 = 2\pi\sqrt{l/g}$

1 Bob immersed in a liquid



$$g_{eff} = g\left(1 - \frac{\rho}{\sigma}\right)$$

bob density σ, *liquid density* $\rho < \sigma$ · *buoyancy only: NON-viscous liquid, drag neglected*

Example

$\sigma = 4\rho$: $g_{eff} = \frac{3}{4}g$, $T = \frac{T_0}{\sqrt{0.75}} = 1.155\,T_0$.

2 Charged bob in a uniform field

$$g_{eff} = g + \frac{qE}{m} \quad (q\vec{E} \text{ down})$$

$$g_{eff} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \quad (\vec{E} \text{ horizontal})$$

uniform $\vec{E}$, *charge* q *on a bob of mass* m, *insulating string* · $q\vec{E}$ *up:* $g - \frac{qE}{m}$

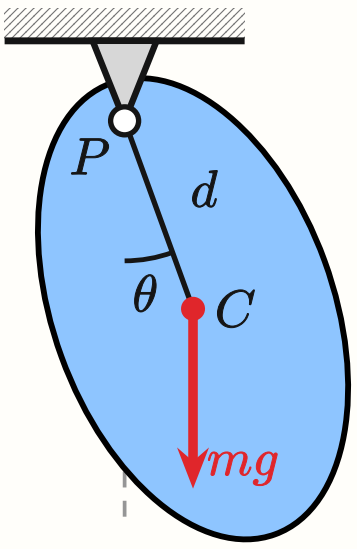
Example

$m = 10\,\text{g}$, $q = 10\,\mu\text{C}$, $\vec{E}$ *downward*, $E = 4900\,\text{N C}^{-1}$:
$\frac{qE}{m} = 4.9\,\text{m s}^{-2}$, $g_{eff} = 14.7\,\text{m s}^{-2}$ $(g = 9.8\,\text{m s}^{-2})$,
$T = \frac{T_0}{\sqrt{1.5}} = 0.816\,T_0$.

Physical (Compound) Pendulum

rigid body · fixed HORIZONTAL axis through the pivot, frictionless · small angle · uniform g

1 Rigid body about a pivot



$$T = 2\pi\sqrt{\frac{I}{mgd}} = 2\pi\sqrt{\frac{k^2 + d^2}{gd}}$$

I about the PIVOT $= m(k^2 + d^2)$ · k: radius of gyration about C · $d = PC$

- *Equivalent length $d + \frac{k^2}{d}$; least T at $d = k$: $T_{min} = 2\pi\sqrt{\frac{2k}{g}}$.*

2 Standard bodies

rod, end: $2\pi\sqrt{\frac{2L}{3g}}$ *ring on a nail:* $2\pi\sqrt{\frac{2R}{g}}$

disc, rim point: $2\pi\sqrt{\frac{3R}{2g}}$

uniform rod, length L · ring / disc, radius R, axis $\perp$ its plane

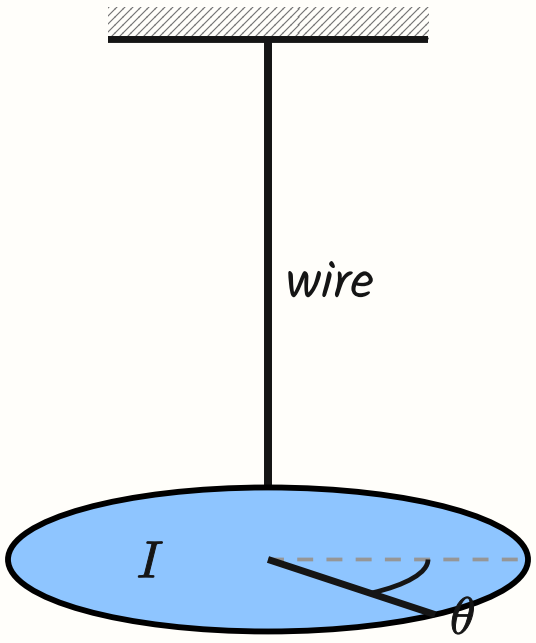
Example

Rod, $L = 1.5\,\mathrm{m}$, end pivot, $g = 9.8\,\mathrm{m\,s^{-2}}$: $T = 2\pi\sqrt{\frac{2\times1.5}{3\times9.8}} = 2.007\,\mathrm{s}$ *(as a $1\,\mathrm{m}$ simple pendulum).*

Torsional Pendulum

wire within the elastic limit · wire mass neglected · clamped rigid support

1 Twisted wire



$$\tau = -C\theta \qquad T = 2\pi\sqrt{\frac{I}{C}}$$

C = torsion constant ($\mathrm{N\,m\,rad^{-1}}$) · I about the WIRE axis · linear for any twist within the elastic limit (no small-angle condition)

$$C = \frac{\pi\eta r^4}{2l}$$

uniform wire: radius r, length l, modulus of rigidity η · $C \propto r^4$: halving r makes T FOUR times

2 Angular SHM

- $I\alpha = -C\theta$ gives $\alpha = -\omega^2\theta$, $\omega = \sqrt{C/I}$; $\theta = \theta_0 \sin(\omega t + \phi)$.

Example

Disc with $I = 0.02\,\mathrm{kg\,m^2}$ on a wire with $C = 0.08\,\mathrm{N\,m\,rad^{-1}}$: $T = 2\pi\sqrt{0.25} = \pi = 3.14\,\mathrm{s}$.

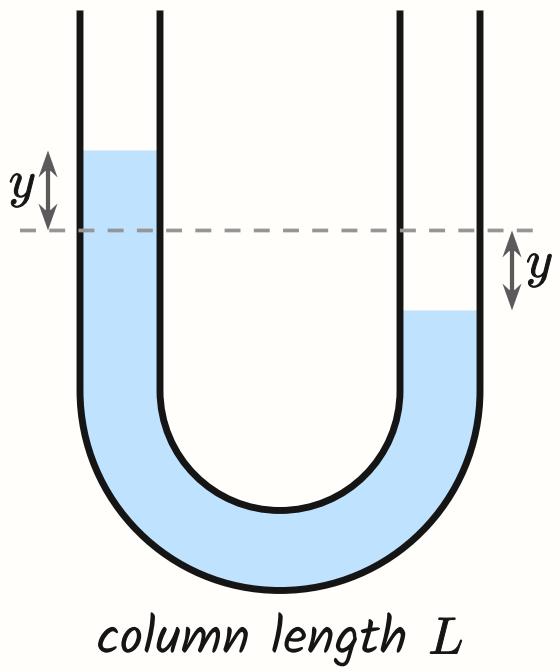
Golden points

★ Same wire, two bodies: $\frac{T_1^2}{T_2^2} = \frac{I_1}{I_2}$, so an unknown I is measured by timing.

U-tube Liquid and Floating Body

NON-viscous liquid · surface tension neglected · uniform g · small displacement

1 Liquid in a U-tube



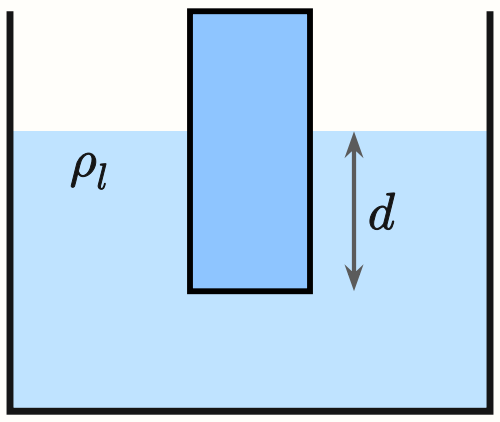
column length L

$$T = 2\pi\sqrt{\frac{L}{2g}}$$

UNIFORM cross-section, vertical arms · L = total length of the liquid column, including the bend

- Level difference $2y$: force $2y\rho gA$ on mass ρAL, $\omega^2 = \frac{2g}{L}$, any density. $L = 0.4\,\text{m}$: $T = 2\pi\sqrt{0.4/19.6} = 0.898\,\text{s}$.

2 Floating body



$$T = 2\pi\sqrt{\frac{m}{\rho_l A g}} = 2\pi\sqrt{\frac{d}{g}}$$

vertical cylinder of uniform cross-section A, floating UPRIGHT · d = depth under the liquid at equilibrium · drag neglected, never fully submerged or out

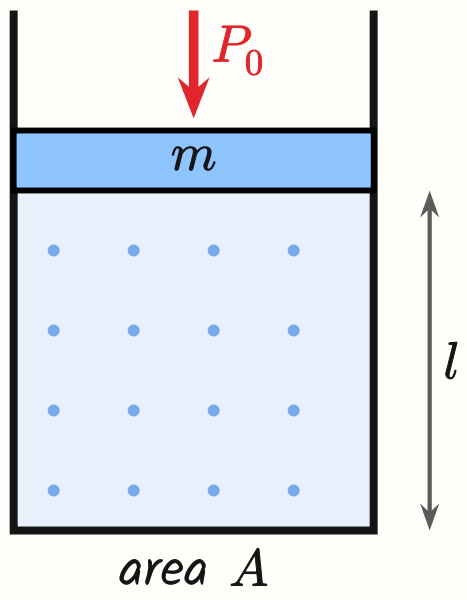
Example

$d = 2.45\,\text{cm}$: $T = 2\pi\sqrt{0.0245/9.8} = 2\pi \times 0.05 = 0.314\,\text{s}$.

Gas Column and Piston

ideal gas · FRICTIONLESS, gas-tight piston · small displacement $y \ll l$

1 Piston on a gas column



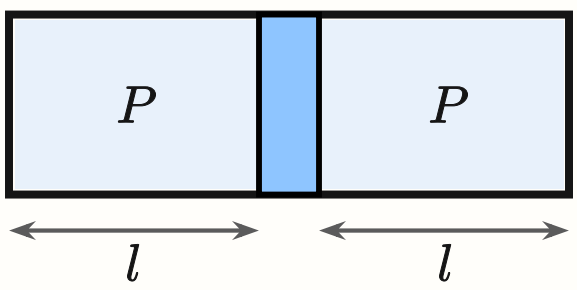
$$T = 2\pi\sqrt{\frac{ml}{PA}} \quad \text{(isothermal)}$$

$$T = 2\pi\sqrt{\frac{ml}{\gamma PA}} \quad \text{(adiabatic)}$$

$P = P_0 + \frac{mg}{A}$ = gas pressure at equilibrium (vertical cylinder) · l = gas column length · $\gamma = C_p/C_v$

- Piston in by y: isothermal $\Delta P = \frac{Py}{l}$, so $k = \frac{PA}{l}$; adiabatic $k = \frac{\gamma PA}{l}$.

2 Piston between two gas columns



$$T = 2\pi\sqrt{\frac{ml}{2PA}}$$

closed horizontal cylinder, equal columns l at pressure P · isothermal · both sides push back: $k = \frac{2PA}{l}$

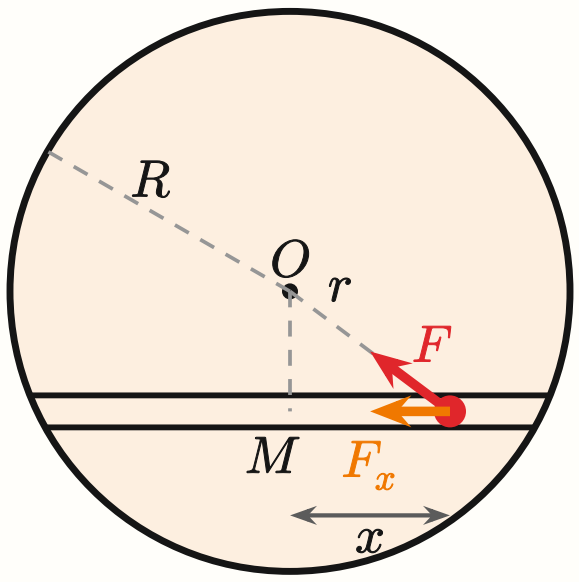
Example

$m = 1\,\text{kg}$, $A = 0.01\,\text{m}^2$, $P = 10^5\,\text{Pa}$, $l = 0.1\,\text{m}$ (one column, isothermal): $k = \frac{PA}{l} = 10^4\,\text{N m}^{-1}$, $T = 2\pi\sqrt{10^{-4}} = 0.0628\,\text{s}$.

Tunnel Through the Earth

UNIFORM-density earth · smooth tunnel, no air · earth's rotation ignored · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$

1 Force inside the earth



$$F = -\frac{mg}{R}r \;\Rightarrow\; F_x = -\frac{mg}{R}x$$

inside a uniform sphere only the mass within radius r pulls: $g(r) = g\frac{r}{R}$ · x from the tunnel's mid-point M

2 Period, any straight tunnel

$$T = 2\pi\sqrt{\frac{R}{g}} = 84.6\,\mathrm{min}$$

$$v_{centre} = \sqrt{gR} = 7.92\,\mathrm{km\,s^{-1}} \quad \textit{(diametral tunnel)}$$

same T for EVERY chord (any length) · end to end in $\frac{T}{2} = 42.3\,\mathrm{min}$ · speed at M: $\omega\times$ half the chord

- *T equals the period of a satellite skimming the surface; dropped from the surface, the body starts at an EXTREME, so the amplitude is half the chord.*

Example

$\sqrt{R/g} = \sqrt{6.4 \times 10^6/9.8} = 808\,\mathrm{s}$;
$T = 2\pi \times 808 = 5077\,\mathrm{s} = 84.6\,\mathrm{min}$; $\sqrt{gR} = 7.92 \times 10^3\,\mathrm{m\,s^{-1}}$.

SHM from Potential Energy

small oscillations about a STABLE equilibrium · one-dimensional motion, conservative force

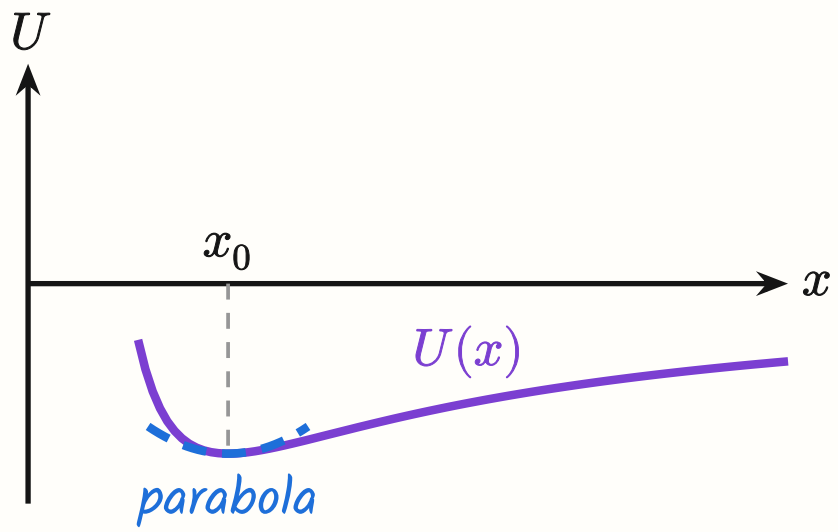
1 The rule

$$F = -\frac{dU}{dx} \qquad U'(x_0) = 0,\ U''(x_0) > 0$$

$$\omega^2 = \frac{U''(x_0)}{m} \qquad T = 2\pi\sqrt{\frac{m}{U''(x_0)}}$$

x_0 = equilibrium (minimum of U) · near x_0: $U \approx U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$, so $k_{eff} = U''(x_0)$

2 Potential well



- Drawn: $U = \frac{a}{x^2} - \frac{b}{x}$ with $a = 1$, $b = 2$: $x_0 = \frac{2a}{b}$, $U''(x_0) = \frac{b^4}{8a^3}$.
- Maximum of U: unstable.
  $U = cx^4$: oscillates, NOT SHM.

3 Worked case

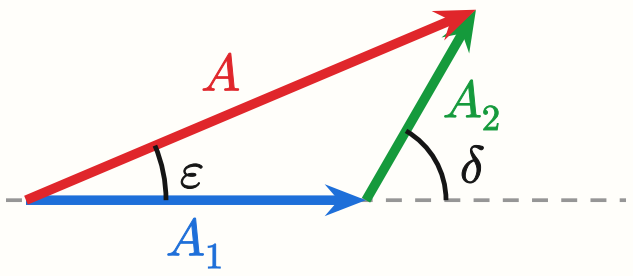
Example

$U = (2x^2 - 4x + 5)\,\mathrm{J}$ (x in m), $m = 0.5\,\mathrm{kg}$: $U' = 4x - 4 = 0$ at $x_0 = 1\,\mathrm{m}$; $U'' = 4\,\mathrm{N\,m^{-1}}$; $\omega = \sqrt{4/0.5} = 2.83\,\mathrm{rad\,s^{-1}}$, $T = 2.22\,\mathrm{s}$.

Superposition: Same Direction

same direction, SAME angular frequency ω

1 Two SHMs add to one SHM



$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\delta}$$

$$\tan\varepsilon = \frac{A_2\sin\delta}{A_1 + A_2\cos\delta}$$

$x_1 = A_1\sin\omega t$, $x_2 = A_2\sin(\omega t + \delta)$ · *result* $x = A\sin(\omega t + \varepsilon)$, *SAME* ω

- *Equal amplitudes:* $A = 2A_1\cos\frac{\delta}{2}$, $\varepsilon = \frac{\delta}{2}$.
- *Phasors add like vectors.* $\delta = 0$: $A_1 + A_2$. $\delta = \pi$: $|A_1 - A_2|$. $\delta = \frac{\pi}{2}$: $\sqrt{A_1^2 + A_2^2}$.

2 Examples

Example

$x = 3\sin\omega t + 4\cos\omega t$: $\delta = \frac{\pi}{2}$, $A = 5$, $\tan\varepsilon = \frac{4}{3}$, $\varepsilon = 53°$:
$x = 5\sin(\omega t + 53°)$.
$A_1 = A_2 = 2\,\text{cm}$, $\delta = \frac{2\pi}{3}$: $A = \sqrt{4 + 4 - 4} = 2\,\text{cm}$.

- *Three equal SHMs with phases* 0, $\frac{2\pi}{3}$, $\frac{4\pi}{3}$: *the phasors close a triangle,* $A = 0$.
- *DIFFERENT* ω: *the sum is not SHM (it is periodic only if* ω_1/ω_2 *is rational; beats in Waves).*

Perpendicular SHMs (Lissajous)

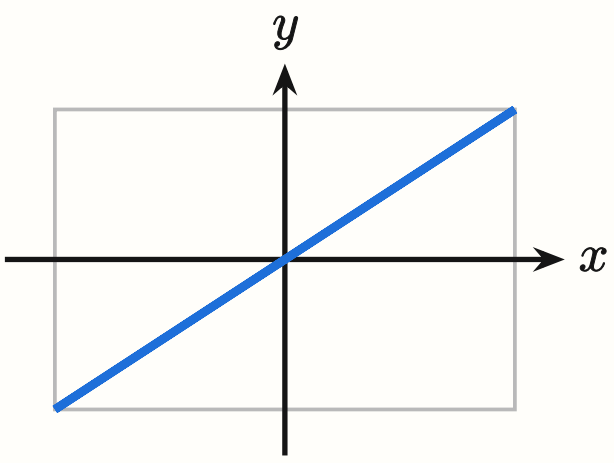
$x = a\sin\omega t$, $y = b\sin(\omega t + \delta)$ · **perpendicular directions, same ω unless stated**

1 Path of the particle

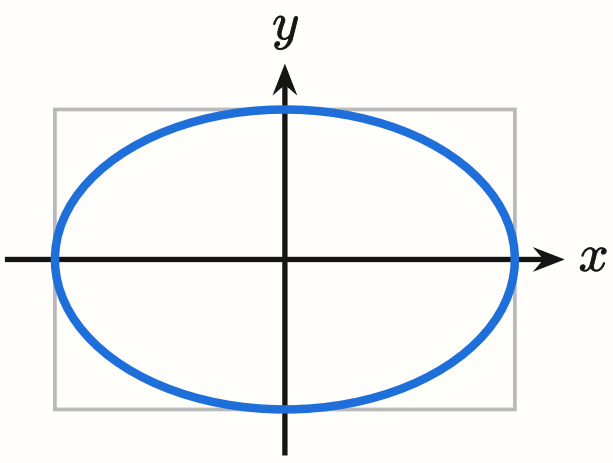
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta$$

eliminate t · in general an ELLIPSE inside the $2a \times 2b$ box

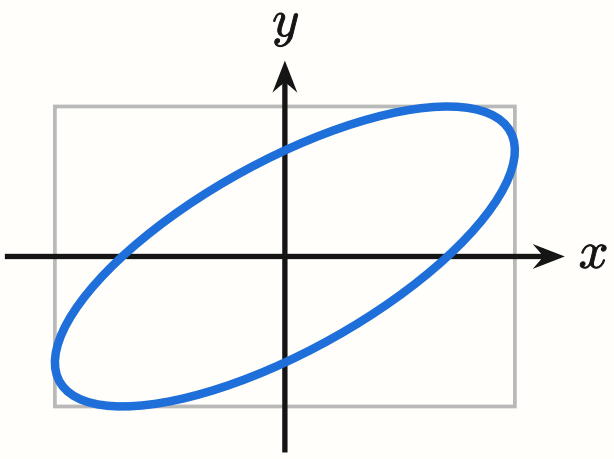
2 Standard shapes



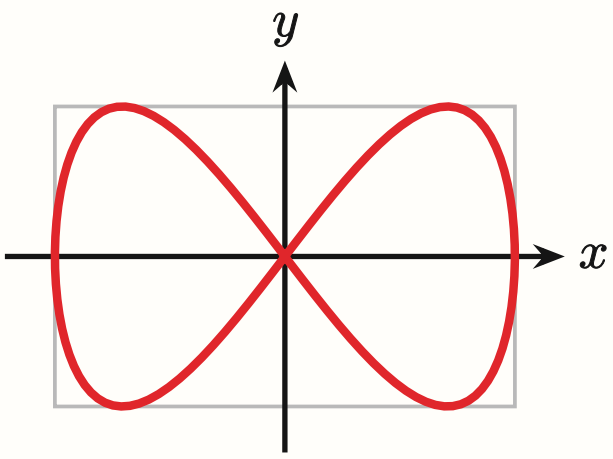
$\delta = 0$: line $y = \frac{b}{a}x$



$\delta = \frac{\pi}{2}$: ellipse



$\delta = \frac{\pi}{4}$: tilted ellipse



$y = b\sin 2\omega t$: figure of eight

- $\delta = \pi$: line $y = -\frac{b}{a}x$. $\delta = \frac{\pi}{2}$, $a = b$: CIRCLE. A line path is an SHM of amplitude $\sqrt{a^2 + b^2}$.

Damped Oscillations

damping force $-bv$ (proportional to velocity) · LIGHT damping

$b < 2\sqrt{km}$ · $\omega = \sqrt{k/m}$

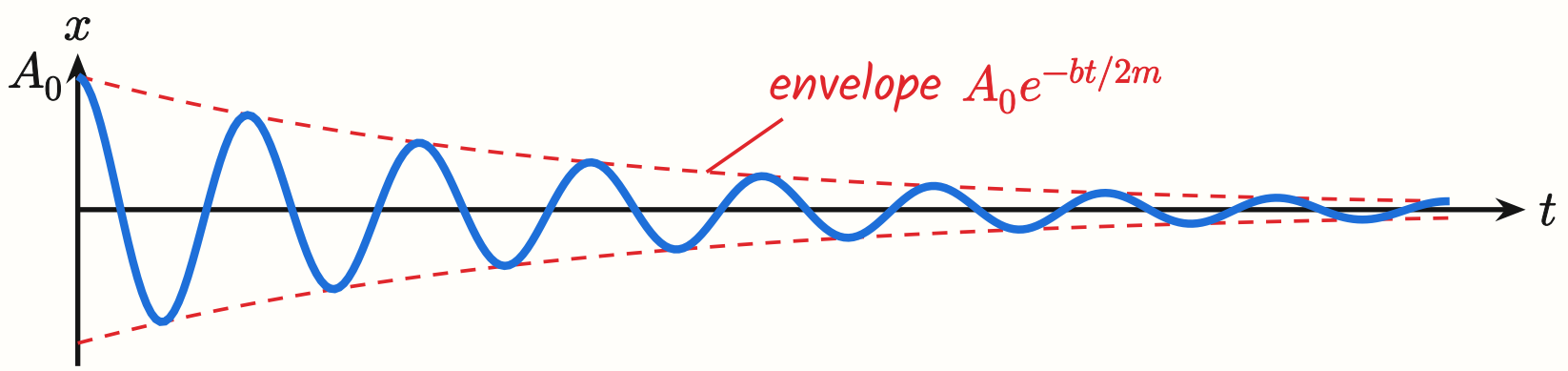
1 Equation and solution

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

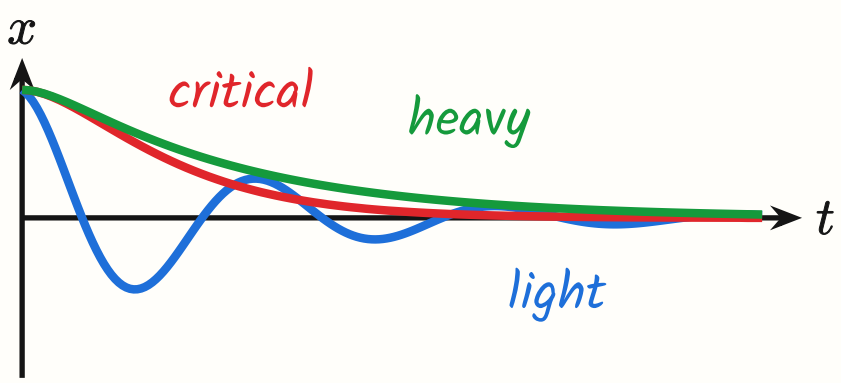
$$x = A_0 e^{-bt/2m}\cos(\omega' t + \phi) \qquad \omega' = \sqrt{\omega^2 - \frac{b^2}{4m^2}}$$

b = damping constant ($\mathrm{kg\,s^{-1}}$) · amplitude $A_0 e^{-bt/2m}$ · $\omega' < \omega$ (slightly slower)

2 The graph



3 Light, critical, heavy damping



- $b < 2\sqrt{km}$: decaying oscillation.
- $b = 2\sqrt{km}$: critical, FASTEST return, no oscillation.
- $b > 2\sqrt{km}$: overdamped, slow.

Damping: Energy and Decay

damping force $-bv$ · LIGHT damping $b \ll 2\sqrt{km}$ · t from the start, amplitude A_0, energy E_0

1 Energy and decay times

$$E = E_0 e^{-bt/m}$$

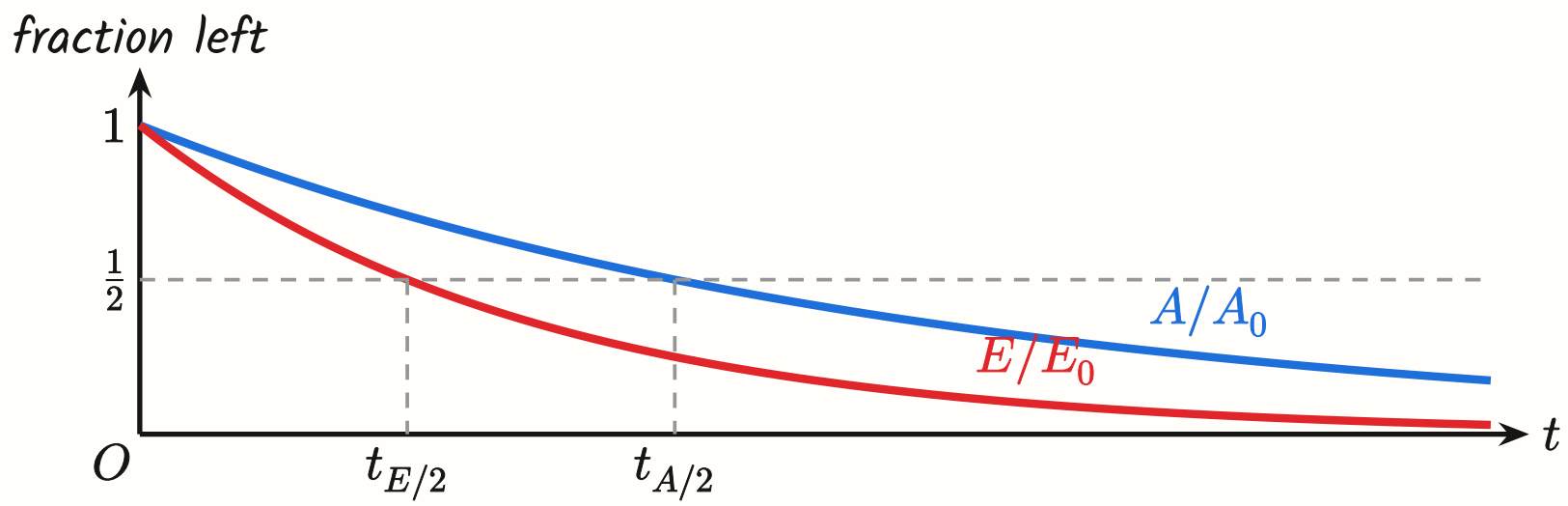
$$t_{A/2} = \frac{2m\ln 2}{b} \qquad t_{E/2} = \frac{m\ln 2}{b}$$

light damping ($b \ll 2\sqrt{km}$), E averaged over one cycle · energy $\propto$ amplitude^2, so it decays TWICE as fast

Example

$m = 0.5\,\text{kg}$, $b = 0.1\,\text{kg s}^{-1}$: $t_{A/2} = \frac{2\times0.5\times0.693}{0.1} = 6.93\,\text{s}$, $t_{E/2} = 3.47\,\text{s}$.

2 Amplitude and energy against t



3 Equal times, equal factors

Example

Amplitude falls to $0.9A_0$ in the first $10\,\text{s}$. After $30\,\text{s}$: $0.9^3 A_0 = 0.729A_0$; *energy* $0.81^3 E_0 = 0.531E_0$.

Forced Oscillations and Resonance

driving force $F_0 \cos \omega_d t$ · damping $-bv$ · $\omega = \sqrt{k/m}$ = natural angular frequency · STEADY state

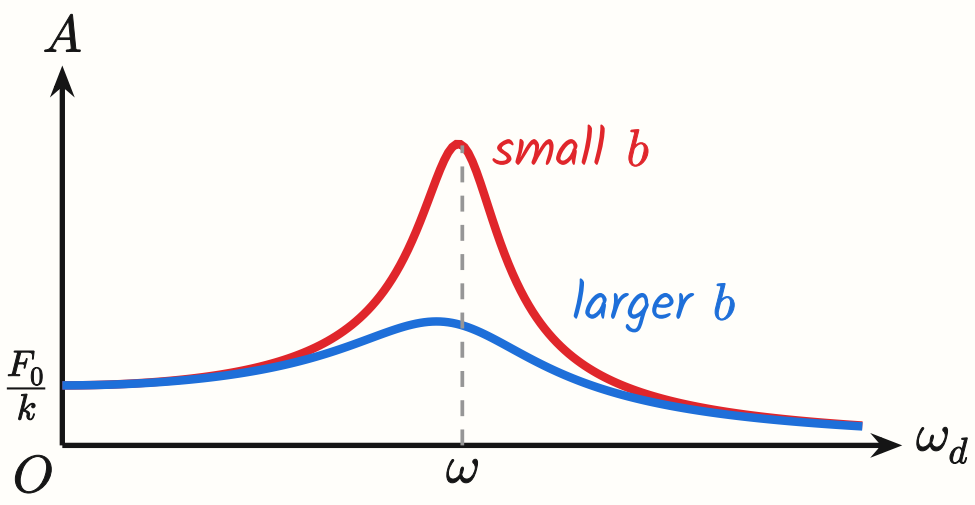
1 Steady-state amplitude

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega_d t$$

$$A = \frac{F_0}{\sqrt{m^2(\omega^2 - \omega_d^2)^2 + b^2\omega_d^2}}$$

after the transient (natural) motion has died out, the body oscillates at the DRIVING frequency ω_d, not at ω

2 Amplitude against driving frequency



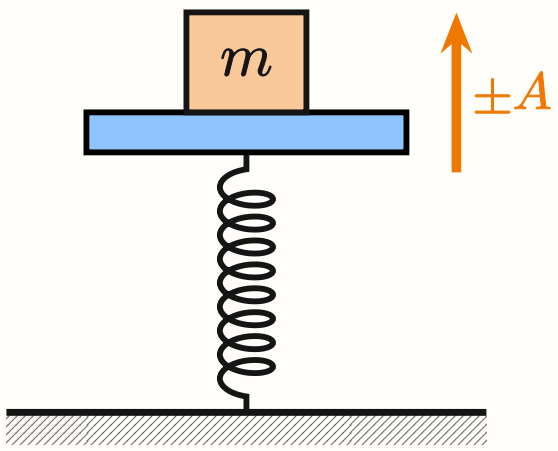
- *RESONANCE: amplitude maximum when $\omega_d \approx \omega$ (light damping).*
- *Smaller b: taller, SHARPER peak; $b \to 0$: $A \to \infty$ at $\omega_d = \omega$.*
- *$\omega_d \to 0$: $A \to \frac{F_0}{k}$; $\omega_d \gg \omega$: $A \to 0$.*

- *At $\omega_d = \omega$: $A = \frac{F_0}{b\omega}$ exactly; the true peak, at $\omega_d = \sqrt{\omega^2 - \frac{b^2}{2m^2}}$, is just below ω (light damping).*
- *Examples: soldiers break step on a bridge; a swing pushed at its own frequency; radio tuning.*

Special Cases (I)

ideal springs · uniform $g = 9.8\,\mathrm{m\,s^{-2}}$ *· small oscillations*

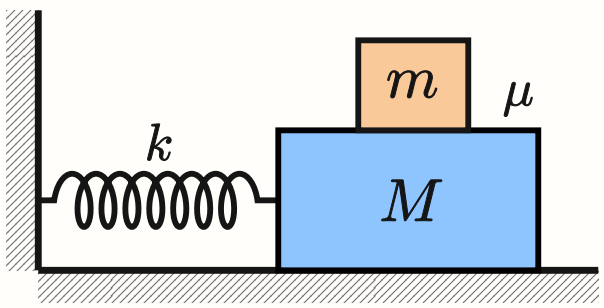
1 Block on a vertical spring platform



$$A_{max} = \frac{g}{\omega^2}$$

block RESTING on a platform in vertical SHM · contact is lost at the TOP if $A\omega^2 > g$ *(normal force would be negative)*

2 Block on a block (friction)



$$A_{max} = \frac{\mu g}{\omega^2} = \frac{\mu g(M+m)}{k}$$

smooth floor, friction coefficient μ *only between the blocks · no slipping if* $A\omega^2 \le \mu g$, $\omega^2 = \frac{k}{M+m}$

3 Pendulum clock: temperature and height

$$\frac{\Delta T}{T} = \tfrac{1}{2}\alpha\,\Delta\theta \qquad \frac{\Delta T}{T} = \frac{h}{R}$$

brass / steel rod, linear expansion coefficient α*, temperature rise* $\Delta\theta$ *· height* $h \ll R$*:* g *falls to* $g(1-\frac{2h}{R})$ *·* T *grows: clock LOSES time*

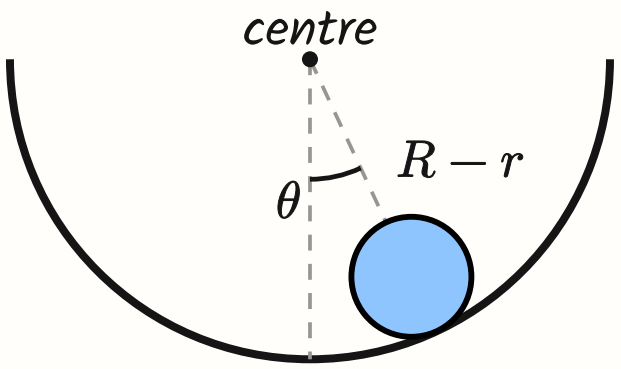
Example

$\alpha = 2\times10^{-5}\,^\circ\mathrm{C}^{-1}$, $\Delta\theta = 10^\circ\mathrm{C}$: $\frac{\Delta T}{T} = 10^{-4}$, *loses* $10^{-4}\times 86400 = 8.64\,\mathrm{s}$ *a day. At* $h = 6.4\,\mathrm{km}$: $\frac{h}{R} = 10^{-3}$, *loses* $86.4\,\mathrm{s}$ *a day.*

Special Cases (2)

uniform g · small oscillations · ideal springs, smooth floor

1 Ball in a smooth / rough bowl



$$T = 2\pi\sqrt{\frac{R-r}{g}}$$

$$T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$$

first: SMOOTH bowl (ball slides) · second: solid sphere ROLLING without slipping · bowl radius R, ball radius r

2 Mass added to an oscillating block

at the mean: $A' = A\sqrt{\dfrac{M}{M+m}}$

at an extreme: $A' = A$

block M on a spring k, smooth floor · m dropped on gently and sticks (momentum conserved at the mean; at an extreme $v = 0$) · both cases: $T' = 2\pi\sqrt{\frac{M+m}{k}}$

Example

$M = 3\,\text{kg}$, $m = 1\,\text{kg}$, $A = 8\,\text{cm}$, *at the mean:*
$A' = 8\sqrt{0.75} = 6.93\,\text{cm}$; T *grows by* $\sqrt{4/3} = 1.155$ *times.*

Standard Results

1 One-line results

System	Result	Condition
SHM	$a = -\omega^2 x$, $v = \omega\sqrt{A^2 - x^2}$	x from equilibrium
Energy	$E = \frac{1}{2}kA^2$	no damping, PE zero at mean
Spring–block	$T = 2\pi\sqrt{m/k}$	ideal massless spring; horizontal, vertical or incline
Series / parallel	$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ / $k_1 + k_2$	ideal springs
Cut spring	kl = constant	uniform spring
Massive spring	$2\pi\sqrt{(m + m_s/3)/k}$	uniform spring, one end fixed
Simple pendulum	$T = 2\pi\sqrt{l/g_{eff}}$	small angle, $\vec{g}_{eff} = \vec{g} - \vec{a}$
Physical	$2\pi\sqrt{I/mgd}$	rigid body, fixed horizontal axis, small angle
Torsional	$2\pi\sqrt{I/C}$	within the elastic limit
U-tube / floating	$2\pi\sqrt{L/2g}$ / $2\pi\sqrt{d/g}$	non-viscous liquid, uniform cross-section
Tunnel	$2\pi\sqrt{R/g}$	uniform earth, smooth tunnel, any chord
From $U(x)$	$\omega^2 = U''(x_0)/m$	small oscillations, stable equilibrium
Damped	$A_0 e^{-bt/2m}$	force $-bv$, light damping $b < 2\sqrt{km}$

Summary

$$a = -\omega^2 x \;\bullet\; E = \tfrac{1}{2}kA^2 \;\bullet\; T = 2\pi\sqrt{\frac{m}{k}} \;\bullet\; T = 2\pi\sqrt{\frac{l}{g}}$$

x from equilibrium · no damping · ideal massless spring · pendulum: small angle

Common Traps

1 Exam traps

Avoid these

- ✗ Every oscillation is SHM (no: SHM needs a LINEAR restoring force, $a = -\omega^2 x$).
- ✗ x measured from the natural length of a vertical spring (no: from the NEW equilibrium).
- ✗ Pendulum T depends on the mass or on a small amplitude (no; only l and g_{eff}).
- ✗ K and U oscillate at f (no: at $2f$, period $\frac{T}{2}$).
- ✗ Average KE over time $= \frac{2}{3}E$ (no: $\frac{E}{2}$; $\frac{2}{3}E$ is the average over POSITION).
- ✗ Cutting a spring in half halves k (no: it DOUBLES k).
- ✗ $\tan\phi = \frac{\omega x_0}{v_0}$ fixes ϕ (check the quadrant with the signs of x_0, v_0).
- ✗ Hollow bob slowly leaking water keeps T (no: the centre of mass falls, then rises; T increases, then decreases).
- ✗ A pendulum oscillates in an orbiting satellite (no: $g_{eff} = 0$; a spring–block still gives $2\pi\sqrt{m/k}$).
- ✗ Resonance at $\omega_d = 2\omega$, or forced motion at the natural ω (no: at $\omega_d \approx \omega$; it moves at ω_d).

Golden points

- ★ First ask: where is equilibrium? what is the restoring force for a small x? then $\omega^2 = k_{eff}/m_{eff}$.

Questions: JEE Main Level (I)

x from equilibrium · ideal massless springs, smooth floor · small angles · $g = 9.8\,\mathrm{m\,s^{-2}}$

Q1 · JEE Main level · MCQ

A particle moves as $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ cm (t in s). Its period and maximum speed: (A) $2\,\mathrm{s}$, $15.7\,\mathrm{cm\,s^{-1}}$ (B) $1\,\mathrm{s}$, $15.7\,\mathrm{cm\,s^{-1}}$ (C) $2\,\mathrm{s}$, $5\,\mathrm{cm\,s^{-1}}$ (D) $\pi\,\mathrm{s}$, $5\,\mathrm{cm\,s^{-1}}$

Solution: $\omega = \pi\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{\omega} = 2\,\mathrm{s}$; $v_{max} = A\omega = 5\pi = 15.7\,\mathrm{cm\,s^{-1}}$.

Answer: (A)

Q2 · JEE Main level · Numerical

A $0.5\,\mathrm{kg}$ block on a smooth horizontal floor is attached to an ideal spring of $k = 50\,\mathrm{N\,m^{-1}}$, pulled $4\,\mathrm{cm}$ from equilibrium and released. (i) Maximum speed (in $\mathrm{m\,s^{-1}}$)? (ii) Maximum acceleration (in $\mathrm{m\,s^{-2}}$)?

Solution: $\omega = \sqrt{50/0.5} = 10\,\mathrm{rad\,s^{-1}}$; $v_{max} = A\omega = 0.04 \times 10 = 0.4\,\mathrm{m\,s^{-1}}$; $a_{max} = A\omega^2 = 0.04 \times 100 = 4\,\mathrm{m\,s^{-2}}$.

Answer: (i) 0.4 (ii) 4

Q3 · JEE Main level · MCQ

A simple pendulum has period $2\,\mathrm{s}$ on the ground (small oscillations). In a lift accelerating UPWARD at $\frac{g}{3}$ its period is: (A) $1.73\,\mathrm{s}$ (B) $2.31\,\mathrm{s}$ (C) $2\,\mathrm{s}$ (D) $1.5\,\mathrm{s}$

Solution: $g_{eff} = g + \frac{g}{3} = \frac{4g}{3}$; $T = \frac{2}{\sqrt{4/3}} = \sqrt{3} = 1.73\,\mathrm{s}$.

Answer: (A)

Questions: JEE Main Level (2)

x from equilibrium · ideal massless springs · small angles, frictionless pivot · PE zero at equilibrium · $g = 9.8\,\mathrm{m\,s^{-2}}$

Q4 · JEE Main level · Numerical

A uniform ideal spring of $k = 600\,\mathrm{N\,m^{-1}}$ is cut into three equal parts. Two of the parts, in parallel, hold a $2\,\mathrm{kg}$ block that oscillates vertically. Period (in s)?

Solution: Each part: $3k = 1800\,\mathrm{N\,m^{-1}}$; parallel: $3600\,\mathrm{N\,m^{-1}}$.

$$T = 2\pi\sqrt{\tfrac{2}{3600}} = 0.148\,\mathrm{s}.$$

Answer: 0.148

Q5 · JEE Main level · MCQ

A particle performs SHM of amplitude A (no damping, PE zero at equilibrium). Its kinetic energy is THREE times its potential energy at $x =$ (A) $\frac{A}{2}$ (B) $\frac{A}{\sqrt{2}}$ (C) $\frac{A}{3}$ (D) $\frac{A}{\sqrt{3}}$

Solution: $\frac{1}{2}k(A^2 - x^2) = 3 \times \frac{1}{2}kx^2$ gives $A^2 = 4x^2$, $x = \frac{A}{2}$.

Answer: (A)

Q6 · JEE Main level · Numerical

A uniform rod of length $0.6\,\mathrm{m}$ swings with small amplitude about a fixed horizontal axis through one end (frictionless pivot). Period (in s)?

Solution: $I = \frac{mL^2}{3}$, $d = \frac{L}{2}$; $T = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{1.2}{29.4}} = 1.27\,\mathrm{s}.$

Answer: 1.27

Questions: JEE Main Level (3)

damping force $-bv$ · NON-viscous liquid, uniform U-tube · $g = 9.8\,\mathrm{m\,s^{-2}}$

Q7 · JEE Main level · MCQ

A $0.25\,\mathrm{kg}$ block on a spring of $k = 25\,\mathrm{N\,m^{-1}}$ has damping constant $b = 0.05\,\mathrm{kg\,s^{-1}}$ (light damping). Time for its amplitude to halve:
(A) $6.93\,\mathrm{s}$ (B) $3.47\,\mathrm{s}$ (C) $13.9\,\mathrm{s}$ (D) $10\,\mathrm{s}$

Solution: $b < 2\sqrt{km} = 5\,\mathrm{kg\,s^{-1}}$. $e^{-bt/2m} = \frac{1}{2}$:

$t = \frac{2m\ln 2}{b} = \frac{0.5\times 0.693}{0.05} = 6.93\,\mathrm{s}.$

Answer: (A)

Q8 · JEE Main level · Numerical

Two SHMs along the same line with the same frequency have amplitudes $3\,\mathrm{cm}$ and $4\,\mathrm{cm}$ and a phase difference $\frac{\pi}{3}$. Resultant amplitude (in cm)?

Solution: $A = \sqrt{9 + 16 + 2\times 3\times 4\times \frac{1}{2}} = \sqrt{37} = 6.08\,\mathrm{cm}.$

Answer: 6.08

Q9 · JEE Main level · Numerical

A U-tube of uniform cross-section holds a liquid column of total length $0.98\,\mathrm{m}$. The liquid is displaced slightly and released. Period (in s)?

Solution: $T = 2\pi\sqrt{\frac{L}{2g}} = 2\pi\sqrt{\frac{0.98}{19.6}} = 2\pi\sqrt{0.05} = 1.40\,\mathrm{s}$, whatever the liquid's density.

Answer: 1.40

Questions: JEE Advanced Level (I)

one-dimensional conservative motion · smooth floor · ideal spring · $g = 9.8\,\mathrm{m\,s^{-2}}$

A1 · JEE Advanced level · Multi-correct

A $0.1\,\mathrm{kg}$ particle moves along x in the potential $U = (5x^2 - 10x + 2)\,\mathrm{J}$ (x in m) with total mechanical energy $2\,\mathrm{J}$. Then: (A) equilibrium is at $x = 1\,\mathrm{m}$ (B) the period is $\frac{\pi}{5}\,\mathrm{s}$ (C) the amplitude is $1\,\mathrm{m}$ (D) the maximum speed is $5\,\mathrm{m\,s^{-1}}$

Solution: $U' = 10x - 10 = 0$ at $x_0 = 1\,\mathrm{m}$; $U'' = 10\,\mathrm{N\,m^{-1}}$, $\omega = \sqrt{10/0.1} = 10\,\mathrm{rad\,s^{-1}}$, $T = \frac{2\pi}{10} = \frac{\pi}{5}\,\mathrm{s}$. $U(x_0) = -3\,\mathrm{J}$, so $\frac{1}{2} \times 10A^2 = 2 - (-3) = 5$: $A = 1\,\mathrm{m}$. $K_{max} = 5\,\mathrm{J}$: $v_{max} = \sqrt{2 \times 5/0.1} = 10\,\mathrm{m\,s^{-1}}$, not 5.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

A $2\,\mathrm{kg}$ block on a smooth floor is attached to a wall by an ideal spring of $k = 200\,\mathrm{N\,m^{-1}}$. A $0.5\,\mathrm{kg}$ block rests on it; the coefficient of static friction between the blocks is 0.6. Largest amplitude (in cm) for which the top block does not slip?

Solution: Together: $\omega^2 = \frac{k}{M+m} = \frac{200}{2.5} = 80\,\mathrm{s^{-2}}$. The top block needs $m\omega^2 A \le \mu m g$: $A_{max} = \frac{\mu g}{\omega^2} = \frac{0.6 \times 9.8}{80} = 0.0735\,\mathrm{m} = 7.35\,\mathrm{cm}$.

Answer: 7.35

Golden points

★ Friction on the top block supplies ITS share of the SHM force: largest at the extremes, $m\omega^2 A$.

Questions: JEE Advanced Level (2)

small angles · uniform-density earth, smooth tunnel, no air, rotation ignored · $R = 6.4 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$

A3 · JEE Advanced level · Matching

A simple pendulum has period T_0 at rest. List I: (P) in a lift accelerating up at $\frac{g}{2}$ (Q) in a lift going down at constant speed (R) in a cart accelerating horizontally at g (S) bob of density twice that of a NON-viscous liquid it is fully immersed in (drag neglected). List II: (1) T_0 (2) $\sqrt{2}\,T_0$ (3) $0.816\,T_0$ (4) $0.841\,T_0$

Solution: $T \propto g_{eff}^{-1/2}$. (P) $1.5g$: $\sqrt{2/3} = 0.816$. (Q) g: T_0. (R) $\sqrt{2}g$: $2^{-1/4} = 0.841$. (S) $g(1-\frac{1}{2}) = \frac{g}{2}$: $\sqrt{2}\,T_0$.

Answer: P-3, Q-1, R-4, S-2

A4 · JEE Advanced level · Numerical

A straight smooth tunnel is dug along a chord whose distance from the earth's centre is $\frac{R}{2}$. A ball is released from rest at one end. (i) Time (in min) to reach the other end? (ii) Maximum speed (in $\mathrm{km\,s^{-1}}$)?

Solution: $T = 2\pi\sqrt{R/g} = 84.6\,\mathrm{min}$; end to end $= \frac{T}{2} = 42.3\,\mathrm{min}$.

Amplitude = half the chord $= \sqrt{R^2 - \frac{R^2}{4}} = \frac{\sqrt{3}}{2}R$;

$v_{max} = \omega A = \frac{\sqrt{3}}{2}\sqrt{gR} = 0.866 \times 7.92 = 6.86\,\mathrm{km\,s^{-1}}$.

Answer: (i) 42.3 (ii) 6.86

Golden points

★ The chord changes the AMPLITUDE (and so the speed), never the period.

Questions: JEE Advanced Level (3)

driving force $F_0\cos\omega_d t$, damping $-bv$ · rigid body, fixed horizontal axis, small angle · $g = 9.8\,\mathrm{m\,s^{-2}$

A5 · JEE Advanced level · Multi-correct

A $0.1\,\mathrm{kg}$ block on an ideal spring of $k = 10\,\mathrm{N\,m^{-1}}$ (damping constant $b = 0.2\,\mathrm{kg\,s^{-1}}$) is driven by $F_0\cos\omega_d t$ with $F_0 = 0.5\,\mathrm{N}$. In the steady state: (A) the natural angular frequency is $10\,\mathrm{rad\,s^{-1}}$ (B) at $\omega_d = 10\,\mathrm{rad\,s^{-1}}$ the amplitude is $0.25\,\mathrm{m}$ (C) the block oscillates at ω_d (D) as $\omega_d \to 0$ the amplitude tends to $0.5\,\mathrm{m}$

Solution: $\omega = \sqrt{10/0.1} = 10\,\mathrm{rad\,s^{-1}}$; $b < 2\sqrt{km} = 2\,\mathrm{kg\,s^{-1}}$. At $\omega_d = \omega$: $A = \frac{F_0}{b\omega} = \frac{0.5}{0.2\times10} = 0.25\,\mathrm{m}$. As $\omega_d \to 0$: $A \to \frac{F_0}{k} = 0.05\,\mathrm{m}$.

Answer: (A), (B), (C)

A6 · JEE Advanced level · Numerical

A uniform disc of radius $0.2\,\mathrm{m}$ oscillates in its own vertical plane about a horizontal axis $\perp$ the disc through a point at distance d from its centre. (i) d (in cm) for the SMALLEST period? (ii) That period (in s)?

Solution: $k^2 = \frac{R^2}{2}$, $T = 2\pi\sqrt{\frac{k^2+d^2}{gd}}$ is least at $d = k = \frac{R}{\sqrt{2}} = 0.141\,\mathrm{m}$.

$T_{min} = 2\pi\sqrt{\frac{2k}{g}} = 2\pi\sqrt{\frac{0.283}{9.8}} = 1.07\,\mathrm{s}$.

Answer: (i) 14.1 **(ii)** 1.07

Golden points

- ★ Steady state, at $\omega_d = \omega$: $A = \frac{F_0}{b\omega}$, set by DAMPING; as $\omega_d \to 0$: $A \to \frac{F_0}{k}$.
- ★ Physical pendulum: $T(d)$ has a minimum at $d = k$; a period between T_{min} and $T(d_{max})$ (d_{max} = farthest pivot on the body from C) comes from two values of d, one each side of k; a longer period comes from one $d < k$.